$$y[n] = \frac{5}{6} y[n-1] - \frac{1}{6} y[n-2] + x[n]$$

$$x[n] = \delta[n] - \frac{1}{3}\delta[n-1]$$

Taking Z transform :-

$$Z\{y[n-k]\} = z^{-k} Y(z)$$

$$Y(z) = \frac{5z^{-1}}{6} Y(z) - \frac{1}{6} z^{-2} Y(z) + X(z)$$

$$Y(z)\left[1 - \frac{5z^{-1}}{6} + \frac{1}{6} z^{-2}\right] = X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 - \frac{5z^{-1}}{6} + \frac{1}{6} z^{-2}}$$

condition for stability $\Rightarrow$ poles of $H(z) \leq 1$

putting denominator $= 0$,

$$\left(1 - \frac{5z^{-1}}{6} - \frac{z^{-2}}{6}\right) = 0$$

$$\left(\frac{3}{3} - \frac{z^{-1}}{3}\right)\left(\frac{2}{2} - \frac{z^{-1}}{2}\right) = 0$$

$$z_1 = \frac{1}{3} \quad ; \quad z_2 = \frac{1}{2}$$

$$z_1 = 0.33 \quad z_2 = 0.5$$

$z_1, z_2 < 1$ (lie in unit circle).

$\therefore$ System is stable

$y[n] \Rightarrow$ depends on $y[n-1]$ $y[n-2]$ & $x(n)$

$x[n] \Rightarrow$ depends on $\delta[n]$ and $\delta[n-1]$

$\therefore$ causal. ("doesn't depend of $> n$")